Dynamic Guideline: WHO-Based Marburg - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Exactly 5 WHO Marburg-related guidelines / technical guidance documents synthesized

#	WHO guideline / technical guidance	Publication / update date identified	Target population	Scope and objectives	Role in this dynamic guideline
1	**Infection prevention and control guideline for Ebola and Marburg disease, August 2023**	**11 August 2023**	Health and care workers, health-facility managers, IPC teams, outbreak responders caring for suspected/confirmed Ebola or Marburg disease	WHO's most up-to-date health-facility IPC recommendations for Ebola and Marburg disease; covers screening, triage, isolation, PPE, hand hygiene, exposure management, environmental cleaning, waste, and safe management of dead bodies	Core IPC guideline; WHO states it uses GRADE and best available evidence, with 11 new recommendations, 10 good practice statements, and 9 carried-forward recommendations WHO IPC guideline
2	**Diagnostic testing for Ebola and Marburg virus diseases: interim guidance, 20 December 2024**	**20 December 2024**; posted by WHO **10 January 2025**	Laboratory personnel, clinicians, health workers, public-health officials, and stakeholders involved in diagnosis/care	Guidance from specimen collection through diagnostic analysis and reporting; rapid testing for suspected living/deceased cases; NAAT/RT-PCR confirmation; repeat testing;	Core diagnostic guidance WHO diagnostic testing guidance

#	WHO guideline / technical guidance	Publication / update date identified	Target population	Scope and objectives	Role in this dynamic guideline
				discharge testing; biosafety	
3	**Marburg virus – Therapeutics prioritisation**	**20 January 2025**	Trialists, ministries of health, ethics committees, outbreak researchers, R&D Blueprint stakeholders	Prioritization of candidate treatments for randomized clinical trials in Marburg virus disease	Treatment-research and trial-readiness guidance WHO therapeutics prioritisation
4	**WHO Technical Advisory Group – candidate vaccine prioritization: summary of the evaluations and recommendations on the four Marburg vaccines**	Listed in WHO materials as **15 November 2025**, with underlying review context from **10 October 2024**	Vaccine developers, regulators, trial sponsors, outbreak-response planners, countries at risk	Prioritization of four viral-vectored Marburg vaccine candidates for possible Phase 1/2/3 evaluation in at-risk populations	Vaccine-research and outbreak-trial guidance WHO TAG-CVP report
5	**Assessment and management of health and care workers with possible occupational exposures to Orthoebolavirus or Orthomarburgvirus: implementation guidance**	Refined with expert review in **October–November 2024**	Health and care workers, occupational-health teams, IPC teams, outbreak coordinators	Exposure-risk categorization and management of health/care workers with possible occupational exposure to Ebola or Marburg viruses; supports the 2023 IPC guideline	Occupational-exposure management guidance WHO occupational exposure guidance PDF

Important user-specified WHO title cross-check

The title **"Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)"** was identified, but it is **not a Marburg guideline**. It is a WHO/HRP case study linked to the **2022 WHO Abortion care guideline**, published **11 December 2025**, describing youth engagement through consultations and youth-led convening WHO youth-engagement case study. It is therefore **not counted** among the exactly 5 Marburg-related WHO guidance documents above. Its methods relevance is that WHO describes youth participation as strengthening dialogue, enriching policy development, and helping guidance reflect affected populations' realities WHO webinar.

1.2 Key Recommendations

GRADE interpretation used in this dynamic report:

- **High**: Further research very unlikely to change confidence.
- **Moderate**: Further research may change confidence.
- **Low**: Further research likely to affect confidence.
- **Very Low**: Estimate/recommendation highly uncertain.
- For rare, high-fatality outbreak diseases, WHO often issues strong recommendations despite low direct trial certainty when benefits are large, harms are acceptable, and feasibility/values strongly favor action.

Clinical / public-health domain	Recommendation	GRADE Evidence Quality	Strength
Case recognition and isolation	Immediately screen, identify, isolate, and manage suspected Marburg virus disease cases using Ebola/Marburg precautions, even when common mimics such as malaria are also plausible.	Low	Strong
Diagnostic testing	Test **all living or deceased persons meeting the suspected-case definition** for Ebola or Marburg virus disease.	Moderate	Strong
Specimen type	Use **whole blood or plasma** for living patients and **oral swab** for deceased individuals.	Moderate	Strong
Confirmatory diagnosis	Use **NAAT/RT-PCR** for confirmation and species identification of Orthomarburgvirus infection.	Moderate	Strong
Early negative test	If a suspected living case tests negative on a blood sample collected **<72 hours after symptom onset**, repeat testing using blood collected **>72 hours after symptom onset**.	Low	Strong
Discharge testing	Before discharge of a confirmed patient from a treatment centre, obtain **two negative blood samples at least 48 hours apart**.	Low	Strong
Standard precautions	Apply standard precautions at all times: hand hygiene, respiratory hygiene, appropriate PPE, safe injections, environmental cleaning, and safe waste handling.	Low	Strong

Clinical / public-health domain	Recommendation	GRADE Evidence Quality	Strength
Transmission-based precautions	For suspected/confirmed MVD care, add contact/body-fluid transmission precautions, including appropriate PPE and hand hygiene according to WHO's "5 moments."	Low	Strong
IPC ring approach	During outbreaks, implement or strengthen the IPC ring approach: rapid IPC assessment, screening, early detection, isolation, facility/household decontamination, and exposure-risk mitigation.	Low	Strong
Environmental cleaning and waste	Safely segregate, collect, transport, store, treat, and dispose of healthcare waste; decontaminate contaminated health-facility and household environments.	Low	Strong
Safe and dignified burials	Manage dead bodies safely and respectfully to prevent exposure to infected blood/body fluids while maintaining community trust.	Low	Strong
Occupational exposure	Assess and manage exposed health/care workers using risk categorization, monitoring, and work restrictions/escalation according to WHO implementation guidance.	Low	Strong
Clinical treatment	Provide early optimized supportive care, including rehydration, electrolyte correction, symptom management, organ-supportive care where available, and treatment of coinfections/complications.	Low	Strong
Antivirals / monoclonal antibodies	Do **not** treat investigational antivirals or monoclonal antibodies as standard of care outside ethically approved clinical trials or monitored emergency-use frameworks.	Low	Strong
Vaccines	Do **not** treat any Marburg vaccine as licensed/standard as of 29 April 2026; deploy investigational vaccines only under approved trial or outbreak-response protocols.	Low	Strong
Surveillance and contact tracing	List and monitor contacts for **21 days** after last exposure, integrate laboratory testing, rapid isolation, risk communication, and community engagement.	Low	Strong
Community engagement	Use community engagement, risk communication, social mobilization, and trusted local platforms to support early care-seeking, safe burials, contact tracing, and IPC adherence.	Low	Strong

WHO states that Marburg outbreak control relies on **case management, surveillance and contact tracing, laboratory services, IPC, safe and dignified burials, and social mobilization** WHO fact sheet. WHO also states that **early supportive care with rehydration and symptomatic treatment improves survival**, but that there are **currently no approved vaccines or antiviral treatments** for Marburg virus disease WHO fact sheet.

1.3 Guideline Development Methodology

Evidence review process

- The **August 2023 WHO IPC guideline** was developed using **GRADE** and **Evidence-to-Decision** methodology and the best available evidence to prevent health-facility transmission while enabling safe care WHO IPC guideline, BMJ summary.
- WHO reports that this IPC guideline includes **11 new recommendations**, **10 new good practice statements**, and **9 carried-forward recommendations** from earlier 2014/2016 IPC guidance WHO IPC guideline.
- The 2024 diagnostic guidance updates earlier Ebola laboratory guidance, expands scope to include Marburg virus disease, and covers the diagnostic pathway from specimen collection to reporting WHO diagnostic testing guidance.
- WHO occupational-exposure implementation guidance states that its exposure-risk categories are based on a systematic review of health/care-worker exposure data for Ebola or Marburg disease and were refined through outbreak feedback, training, and expert review WHO occupational exposure guidance PDF.

Consensus method

- WHO IPC guidance was produced by a guideline development group of filovirus IPC experts supported by the WHO secretariat; a patient representative who was both an Ebola survivor and health/care worker participated in the guideline development group BMJ summary.
- The guideline development group prioritized **13 key questions** and **5 background questions** for the IPC update Africa Newsroom/WHO.
- Vaccine and therapeutics prioritization guidance reflects WHO R&D Blueprint and Technical Advisory Group processes for selecting candidates suitable for randomized or adaptive outbreak evaluation WHO therapeutics prioritisation, WHO TAG-CVP report.

Conflict of interest management

- The accessible summaries confirm WHO guideline-development methods and expert-group processes but do **not** enumerate individual declarations of interest for each member. This dynamic report therefore treats WHO's published methodology as the authoritative process record and does not infer undisclosed conflicts beyond the source descriptions.

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search strategy

Evidence reviewed after the August 2023 WHO IPC guideline release included WHO guidance pages, WHO Disease Outbreak News, WHO R&D Blueprint materials, ECDC and CDC resources, systematic reviews, peer-reviewed trial publications, outbreak reports, and clinical-trial registry records. Example targeted queries included:

- `Marburg virus disease diagnosis treatment vaccine trial 2026 WHO CDC ECDC`
- `WHO Ebola Marburg diagnostic testing interim guidance 20 December 2024 RT-PCR`
- `Marburg vaccine Phase 1 Phase 2 Sabin IAVI 2026`
- `WHO Marburg therapeutics vaccine landscape 2025 2026`
- `ClinicalTrials.gov Marburg cAd3 Sabin IAVI Rwanda Uganda Kenya United States`

Databases and sources searched

Source category	Sources used
WHO official guidance	WHO Marburg topic page, WHO fact sheet, WHO IPC guideline, WHO diagnostic testing interim guidance, WHO therapeutics prioritization, WHO TAG vaccine-prioritization report, WHO outbreak pages for Rwanda/Tanzania/Ethiopia
Public-health authorities	ECDC Marburg factsheet, CDC Emerging Infectious Diseases
Peer-reviewed literature / reviews	Lancet Infectious Diseases systematic review summary, BMJ/PMC WHO IPC summary, Phase 1 cAd3-Marburg vaccine publication
Clinical-trial sources	ClinicalTrials.gov records for cAd3-Marburg, Sabin trial/program reports, IAVI trial announcement
Outbreak and research reporting	Sabin, IAVI, CIDRAP, RCCE Collective

Inclusion criteria

- Evidence dated after **11 August 2023** and available by **29 April 2026**.
- Human Marburg virus disease diagnosis, treatment, IPC, outbreak management, vaccine, therapeutics, or surveillance evidence.
- WHO documents and authoritative public-health or peer-reviewed sources.
- Clinical trials or outbreak-response trials involving Marburg vaccines or therapeutics.

Exclusion criteria

- Ebola-only vaccine or therapeutic recommendations not applicable to Marburg.
- Market-only projections without clinical or public-health evidence, except as context.
- Non-WHO youth-engagement guidance not directly relevant to Marburg recommendations.

2.2 Key New Studies and Evidence

Study / evidence source	Design / evidence type	Key findings	GRADE Quality
WHO **Diagnostic testing for Ebola and Marburg virus diseases: interim guidance, 20 December 2024**	WHO interim diagnostic guidance	Test all suspected living/deceased cases; use whole blood/plasma in living patients and oral swab in deceased individuals; confirm by NAAT/RT-PCR; repeat early negative tests after 72 hours; use two negative blood samples ≥48 hours apart before discharge WHO diagnostic testing guidance, ReliefWeb summary	Moderate for NAAT diagnosis; Low for repeat/discharge intervals
WHO Ethiopia Disease Outbreak News, 2026	Outbreak report	Ethiopia outbreak: 14 confirmed cases and 9 confirmed-case deaths as of 25 January 2026; 857 contacts completed 21-day follow-up; 3,800 samples tested by 5 January 2026 WHO Ethiopia outbreak notice	Low
WHO Tanzania / Ethiopia outbreak IPC notices	Outbreak operational guidance	Reinforced 2023 WHO IPC guideline; emphasized IPC ring approach, rapid IPC assessment, decontamination, screening, isolation, PPE, hand hygiene, and safe waste management WHO Tanzania outbreak notice, WHO Ethiopia outbreak notice	Low
WHO Rwanda 2024 outbreak page	Outbreak operational report	WHO supported Rwanda with diagnosis, IPC, patient care, contact tracing, early detection, community engagement, risk communication, and clinical-trial readiness for investigational products WHO Rwanda outbreak page	Low
Lancet Infectious Diseases systematic	Systematic review; PubMed/Web of	Summarized 478 reported cases and 385 deaths from	Low to Moderate for epidemiology; Very Low

Study / evidence source	Design / evidence type	Key findings	GRADE Quality
review of MVD outbreaks, models, and parameters	Science through 31 March 2023; PROSPERO registered	historical MVD outbreaks; highlighted need to improve understanding after 2023 Equatorial Guinea and Tanzania outbreaks ScienceDirect	for intervention effects
CDC Emerging Infectious Diseases genomic report on Tanzania 2023/2025 strains	Genomic/outbreak analysis	Both 2023 and 2025 Tanzania outbreaks occurred in Kagera; local ecology included Egyptian rousette bat habitats, caves, and mining sites CDC EID	Low
Sabin cAd3-Marburg Phase 1 publication	First-in-human Phase 1, open-label, dose-escalation vaccine study	cAd3-Marburg vaccine was safe and well tolerated in healthy U.S. adults and induced durable glycoprotein-specific antibody response PMC Phase 1 study	Low
Sabin Phase 2 program: Uganda, Kenya, U.S., Rwanda, Ethiopia	Phase 2 randomized and outbreak-response vaccine trials	Phase 2 trials in Uganda/Kenya and U.S.; Rwanda outbreak trial enrolled >1,700 participants; Ethiopia outbreak-response trial planned/initiated for high-risk health/front-line workers and contacts Sabin Phase 2 update, Sabin Ethiopia update	Low for safety/immunogenicity; Very Low for efficacy
ClinicalTrials.gov NCT03475056	Phase 1 cAd3-Marburg vaccine trial	Phase 1 interventional trial in healthy adults at WRAIR Clinical Trials Center, Silver Spring, Maryland ClinicalTrials.gov NCT03475056	Low
ClinicalTrials.gov NCT05817422	Phase 2 randomized controlled vaccine trial	Kenya/Uganda trial of cAd3-Marburg vaccine; healthy adults aged 18–70; sponsor Sabin, collaborator BARDA ClinicalTrials.gov NCT05817422	Low
IAVI rVSVΔG-MARV-GP Phase 1 trial announcement	First-in-human Phase 1 vaccine trial announcement	First participants vaccinated in U.S. Phase 1 trial on 6 April 2026; intended to generate foundational human safety	Very Low to Low

Study / evidence source	Design / evidence type	Key findings	GRADE Quality
		and immunogenicity data IAVI press release	
WHO TAG vaccine-prioritization report	WHO expert-prioritization evidence synthesis	Reviewed four Marburg vaccine candidates: two replicating VSV-vectored and two non-replicating chimpanzee adenovirus-vectored candidates encoding Marburg glycoprotein WHO TAG-CVP report	Low
Rwanda Marburg therapeutic trial report	Outbreak clinical-trial report	Rwanda and WHO partners launched trials of remdesivir and MBP091 monoclonal antibody for MVD treatment; no approved treatments/vaccines at time of reporting CIDRAP	Very Low
ECDC Marburg factsheet	Public-health technical factsheet	Supports RT-PCR/ELISA biosafety distinctions; non-inactivated samples require BSL-3 for RT-PCR/ELISA and BSL-4 for virus isolation; inactivated samples may be tested at BSL-2 ECDC	Low

2.3 Clinically Actionable Synthesis

Diagnosis

1. **Do not rely on clinical features alone.** Marburg virus disease is difficult to distinguish clinically from malaria, typhoid fever, shigellosis, meningitis, and other viral haemorrhagic fevers WHO Marburg health topic.
2. **Test every suspected case.** WHO diagnostic guidance states that rapid diagnosis is critical for timely care and stopping transmission WHO diagnostic testing guidance.
3. **Use NAAT/RT-PCR as the confirmatory method.** Laboratory confirmation and species identification should be performed using nucleic acid amplification testing ReliefWeb summary of WHO diagnostic guidance.
4. **Specimens:** whole blood or plasma for living patients; oral swab for deceased individuals ReliefWeb summary.
5. **Repeat testing:** if blood was drawn <72 hours after symptom onset and is negative, repeat after 72 hours because early viral load may be below detection BMJ Best Practice.

6. **Discharge testing:** two negative blood samples at least 48 hours apart before discharge from a treatment centre BMJ Best Practice.

Treatment and clinical management

- **No approved Marburg antiviral or vaccine** was identified in WHO sources as of 29 April 2026 WHO fact sheet.
- **Supportive care remains standard management:** early rehydration, electrolyte correction, symptom control, management of shock/bleeding/organ dysfunction where resources allow, and treatment of coinfections or alternative diagnoses.
- **Investigational treatments:** remdesivir and MBP091 were reported in Rwanda treatment trials, but evidence remains insufficient for routine standard-of-care use outside approved trial or monitored protocols CIDRAP.
- Ebola-specific monoclonal antibodies and vaccines should **not** be extrapolated to Marburg; WHO's Marburg materials state no approved Marburg vaccines or antiviral treatments currently exist WHO fact sheet.

Infection prevention and control

- Use **standard precautions for all patients** and add **transmission-based precautions** for suspected/confirmed MVD.
- Implement screening, triage, rapid isolation, PPE, hand hygiene, environmental cleaning, decontamination, waste management, occupational-exposure management, and safe/decent burials.
- WHO outbreak notices specifically emphasize the **IPC ring approach**, rapid IPC assessment, decontamination of health facilities and households, early detection, screening, and isolation WHO Tanzania outbreak notice, WHO Ethiopia outbreak notice.

Vaccines

- **No licensed Marburg vaccine** as of the evidence cutoff.
- Sabin's cAd3-Marburg vaccine has Phase 1 safety/immunogenicity data and Phase 2 studies/outbreak-response trials in Africa and the U.S. PMC Phase 1 study, Sabin Phase 2 update.
- IAVI began a U.S. Phase 1 trial of a single-dose rVSVΔG-MARV-GP vaccine candidate in April 2026 IAVI press release.
- Use of investigational vaccines should occur only in ethically approved trials or outbreak-response protocols with regulatory and community oversight.

2.4 Evidence Gaps and Future Research Needs

Evidence gap	Current state	Research priority	GRADE certainty of current evidence
Therapeutic efficacy	No approved treatment; remdesivir and MBP091 under outbreak-trial evaluation	Adaptive randomized trials during outbreaks; standardized endpoints including mortality, viral clearance, safety	Very Low

Evidence gap	Current state	Research priority	GRADE certainty of current evidence
Vaccine efficacy	Phase 1/2 safety and immunogenicity evidence emerging; no licensed vaccine	Ring-vaccination/outbreak trials, immune correlates of protection, durability, special populations	Low for safety; Very Low for efficacy
Optimal supportive-care package	Supportive care improves survival, but specific resource-stratified bundles are not well quantified	Comparative implementation studies of fluids, electrolytes, organ support, transfusion strategies, coinfection management	Low
Diagnostic performance in field settings	NAAT/RT-PCR recommended; field logistics and early false negatives remain challenges	Field diagnostic accuracy studies, sample stability, point-of-care NAAT, antigen test validation for Marburg	Moderate for NAAT principle; Low for operational performance
Occupational exposure risk	WHO exposure guidance exists; direct data limited	Prospective exposure registries, PPE breach analysis, post-exposure protocols	Low
IPC implementation in resource-limited settings	Strong recommendations but implementation barriers persist	Pragmatic IPC implementation trials, WASH infrastructure studies, PPE usability/safety studies	Low
Community engagement	WHO emphasizes community engagement; youth groups and local platforms may be useful	Co-designed RCCE interventions, trust-building, safe burial acceptability, misinformation response	Low
Transmission ecology	Fruit bats are recognized reservoirs; outbreak spillover prediction remains limited	One Health surveillance, genomic epidemiology, bat-human interface studies	Low

2.5 Updated Recommendations Based on New Evidence

Original WHO-based recommendation	New evidence impact through 29 April 2026	Updated recommendation
Apply WHO 2023 IPC guideline for all suspected/confirmed MVD care	Tanzania and Ethiopia outbreak notices operationalized IPC ring approach, screening, isolation, decontamination, PPE, hand hygiene, and waste management	**Unchanged; strengthened operationally.** Facilities should pre-position IPC/WASH assessment, PPE, waste, isolation, and decontamination capacity before outbreaks.
Use rapid laboratory confirmation for suspected cases	WHO 2024 diagnostic guidance provides more specific testing pathway and specimen recommendations	**Updated.** Test all suspected living/deceased cases; use whole blood/plasma or oral swabs; confirm by NAAT/RT-PCR; repeat early negatives after 72 hours.
Provide supportive care; no approved antiviral treatment	Rwanda therapeutic trials began evaluating remdesivir and MBP091, but no efficacy results establish standard use	**Unchanged.** Supportive care remains standard; investigational therapeutics only in approved trials or monitored emergency frameworks.
No approved Marburg vaccine	Sabin Phase 2 and IAVI Phase 1 vaccine trials show active progress but no licensed product	**Unchanged.** No vaccine is standard of care; investigational deployment only under approved protocols.
Conduct surveillance and contact tracing	Ethiopia outbreak documented 857 contacts completing 21 days of follow-up	**Unchanged; operationally reinforced.** Maintain 21-day contact follow-up, rapid testing, isolation, and community engagement.
Use safe and dignified burials	WHO continues to list safe burials as a core outbreak-control intervention	**Unchanged.** Burial teams require PPE, trained protocols, family/community communication, and cultural adaptation.
Manage occupational exposures	WHO implementation guidance refined exposure-risk categorization and management	**Updated.** All facilities should establish exposure-reporting, risk-categorization, monitoring, and work-restriction pathways before patient care begins.

2.6 Final Dynamic Guideline Recommendations

Recommendation	Actionable implementation	GRADE Evidence Quality	Strength
Establish immediate triage and isolation for	Screen febrile/compatible illness with exposure risk; isolate suspected cases; alert public health	Low	Strong

Recommendation	Actionable implementation	GRADE Evidence Quality	Strength
suspected MVD	and laboratory networks		
Confirm diagnosis with NAAT/RT-PCR	Collect whole blood/plasma from living patients; oral swab from deceased; use risk-based biosafety	Moderate	Strong
Repeat early negative tests	If first sample was <72 hours after symptom onset, maintain precautions and repeat after 72 hours	Low	Strong
Use discharge testing	Require two negative blood samples ≥48 hours apart before discharge from treatment centre	Low	Strong
Provide optimized supportive care	Oral/IV rehydration, electrolytes, glucose, symptom control, shock/bleeding management, organ support where feasible, empiric management of common coinfections as clinically indicated	Low	Strong
Avoid unproven routine therapeutics	Use remdesivir, MBP091, or other candidates only through approved trial/monitored frameworks	Very Low	Strong
Apply standard + transmission-based precautions	PPE, hand hygiene, safe injections, respiratory hygiene, environmental cleaning, waste management, and exposure prevention	Low	Strong
Implement IPC ring approach	Rapid IPC assessment, isolate cases, decontaminate facilities/households, protect contacts and health workers	Low	Strong
Manage healthcare waste safely	Segregate, collect, transport, store, treat, and dispose of waste under national/WHO guidance	Low	Strong
Protect health and care workers	Training, PPE observers/buddies, occupational exposure reporting, risk categorization, monitoring, and psychological support	Low	Strong
Conduct contact tracing for 21 days	Daily monitoring after last exposure; rapid testing/isolation if symptoms develop	Low	Strong
Use safe and dignified burials	Trained burial teams, PPE, body handling precautions, family communication, culturally sensitive practice	Low	Strong
Use investigational vaccines only under protocol	Deploy Sabin/IAVI/other candidates only through ethical, regulatory, and community-approved trial or outbreak-response mechanisms	Low for safety; Very Low for efficacy	Strong

Recommendation	Actionable implementation	GRADE Evidence Quality	Strength
Maintain community engagement	Partner with trusted leaders, community groups, youth groups, faith organizations, radio, and survivor networks to support adherence and trust	Low	Strong

Overall conclusion: As of **29 April 2026**, WHO-based Marburg management remains anchored in **rapid recognition, laboratory confirmation, strict IPC, supportive care, contact tracing, safe burials, occupational protection, and community engagement**. Diagnostics have advanced through WHO's December 2024 interim guidance, while vaccines and therapeutics remain investigational despite active Phase 1/2 and outbreak-response trials.